

Vedant Kulkarni

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EDUCATION

- Indian Institute of Information Technology, Lucknow** India
Bachelor of Technology — Computer Science and Business; CGPA: 8.57 Aug-2024 – May-2028

SKILLS

- **Programming Languages:** Go, C++, JavaScript, TypeScript, Python
- **Web Development:** Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, WebSockets, JWT, Redis
- **Databases:** PostgreSQL, MongoDB
- **Cloud& DevOps:** Git, Docker, Vercel, Render
- **CS Fundamentals:** Data Structures and Algorithms, Database Management Systems, Object Oriented Programming, Operating System

TECHNICAL PROJECTS

- **Sentinel — Distributed Rate Limiting System (Go) — [GitHub]:**
Tech Stack: Go, Redis, REST APIs, Goroutines, Mutex/Atomic Operations
 - * Architected and implemented a distributed rate limiting system in Go using Token Bucket and Sliding Window algorithms, enforcing consistent request limits across horizontally scaled service instances.
 - * Designed a high-concurrency, in-memory rate limiting engine leveraging goroutines and mutex/atomic operations to ensure thread-safe, low-latency request processing.
 - * Implemented Redis-backed distributed coordination and HTTP middleware to maintain global rate-limit consistency and enable fail-fast per-client request rejection with minimal performance overhead.
- **BawarchiAI — AI-Driven Food Waste Reduction Platform — [GitHub]:**
Tech Stack: Next.js, TypeScript, shadcn/ui, Go, PostgreSQL, Python, XGBoost, REST APIs, Docker
 - * Built an end-to-end AI-powered platform for institutional kitchens, forecasting real food demand using historical and contextual data to identify surplus food in advance.
 - * Developed a scalable Golang backend with PostgreSQL and a containerized Python ML service using XGBoost, supporting kitchen data ingestion, surplus detection, NGO redistribution workflows, and model retraining.
 - * Enabled real-time surplus acceptance and impact tracking through role-based dashboards for kitchens and NGOs, facilitating food redistribution and monitoring outcomes such as food saved, meals served, and CO₂ reduction.
- **SlayerConnect — Full-Stack Real-Time Messaging System — [GitHub] [Live]:**
Tech Stack: Node.js, Express.js, Socket.IO, MongoDB, JavaScript, REST APIs, WebSockets, JWT, Cloudinary
 - * Built a full-stack real-time messaging platform using Socket.IO (WebSockets) to support low-latency, bi-directional communication between concurrent users.
 - * Designed an event-driven Node.js backend handling user authentication, socket connection lifecycle, message delivery, and real-time notifications using JWT-based security.
 - * Optimized system performance and reliability by eliminating N+1 queries with MongoDB aggregation, enabling O(1) message routing via socket-to-user mapping, and enforcing secure, idempotent message state updates.

ACHIEVEMENTS

- Achieved the title of **Pupil** at Codeforces Round 1059 (Div. 3) — **Global Rank 4320** Ranklist Link
- **Global Rank 3530** at Codeforces Pinely Round 5 (Div. 1 + Div. 2) Ranklist Link
- **Global Rank 471** at CodeChef Starters 209 Contest Link
- **CodeChef Profile** Profile
- **Codeforces Profile** Profile

POSITION OF RESPONSIBILITY

- Member, Web Wing — Axios (Technical Club), IIIT Lucknow Sept-2025 – Present
- Member, Seraphim (Social Awareness Club), IIIT Lucknow Sept-2024 – Present
- Member, Zephyr (Dance Society), IIIT Lucknow Sept-2024 – Present